Presentation 5

- Links to presentation(s) and code(s) on GitHub

Presentation:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/CallOutcomesFinal_9Dec_EDA2.pdf

Video:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/VideoLink_Dec9_EDASimple2.txt

Code:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/CallOutcomeTagging_9Dec_Vivienne.ipynb

- What did you do?

Since the call_type_code for tagging the calls was in Python, and our group had run into issues capturing all outcomes for each menu with the previous R code, we opted to use the call_type_code as our new starting point and reimplement the outcome tagging in Python. For this, I first found the last non-null activity for each contact session ID and added it as a column to the dataframe. To add the outcome and subtype for each call, I used a mix of two approaches since some of the final activities had multiple possible outcomes. For activities with only one outcome, our group created dictionaries for each activity and its outcome or subtype, and for activities with more than one outcome, I created a list of the conditions and corresponding outcomes and subtypes. Then, I created a function to assign the outcome to each row which started by checking if the conditions for the multiple outcomes were met using np.select then used the dictionaries to map the relevant outcomes to the activity for each call. I used generative AI to help with the implementation of this function to ensure that it could consider both types of cases in one function. Next, using a similar approach as for adding the final activity, I added the final menu as a column to the dataframe based on each contact session ID's final EP Name. However, I knew from our previous presentation that I needed to account for cases where the final EP Name is courtesy callback, queues, or closed queues so I created a function to update the final menu value if the EP Name was one of these values. This function ensured the dataframe was sorted by time so the steps of the call would be in sequential order then it filtered to exclude EP Names that don't reflect the actual final menu, and then grouped by the contact session ID to find the last EP Name of interest. I created a dataframe that only included one instance of each contact session ID and used this to find the outcomes for the overall tag and caller type. To drill down into the outcomes for the specific menus for the non-senior legal issue, I filtered the dataframe to only include rows with this as the caller type. Finally, I used groupby to find the outcomes for each menu and then created a secondary aggregation that also included the subtype of outcome for the menus.

- How does it help the project?

This helps the project by providing further insight into the proportion of positive, neutral, and negative outcomes for the different categories of calls, and hopefully the more detailed look at subtype of outcomes and the specific legal menus can help highlight issues that require more attention.

- Issues faced (if any)

The major issues faced in this analysis was finding a way to handle the mix of outcomes that could be directly mapped to the final activity versus the activites with multiple cases, and ensuring that the correct menu was assigned to each call. I believe that these issues were resolved tentatively so they are discussed further in the 'Issues resolved' section. However, in looking at the classifications and aggregations that these approaches created there were a few questions that arose. Firstly, there are calls that are tagged as non-senior legal issues, but their EP Name and therefore, final menu is Main Menu or Pre-Legal Seniors. In just initially inspecting these, it seems that these calls tend to be ones where the caller has a few steps in different areas of the legal menu but ultimately returns to the main menu or enters the seniors menu. I thought that these should be considered Main Menu or Seniors Menu since that seems to be where they terminate, but I was unsure if there was a rationale for including them with the non-seniors legal issues or if somewhere in tagging the outcomes, there had been a conflict or they had somehow been reclassified (see examples in code). Additionally, for Other Legal Menu, all of the outcomes were neutral, though they were split between 'LAC does not serve' and 'Unknown' (default) in terms of subtype. In looking at examples, the small sample was consistent with this, but overall it seemed implausible so I believe there might be a problem with the combination of the dictionary value for OtherLegalMenu and the conditions that are meant to address all of the possible cases not covering all the potential outcomes. Finally, since EP Name was used as the approximation for the final menu, there is not a distinction between Legal Menu 1 and 2 at this point. I believe from previous analysis we know that the majority of calls do end up passing through Legal Menu 1 to Legal Menu 2, but seeing as abandonments are such a large proportion of the outcomes for Legal Menu, I think it would be helpful to be able to identify at which point within the menus the calls are being abandoned since it would indicate different issues (e.g. Legal Menu 1, caller hasn't heard the legal issues served so maybe they decide they don't have time to call whereas Legal Menu 2 might indicate that they can't make a decision based on the options or they realize LAC does not have an option for their legal issue).

- Attempts to resolve issues (if any)
- Issues resolved (if any)

Initially, the outcomes with one outcome and multiple outcomes were in two separate functions which created the risk of overwriting the previous tagging so I ended up combining these two functions into one. Additionally, there was the challenge of finding a scalable way of finding the last relevant menu as many calls that end on courtesy callback were previously in a queue so simply going to the previous EP was not feasible, and this led to the approach of filtering out the rows with the non-target EP names and then finding the last target EP from each row.

- Next Steps

To continue this analysis, possible steps would be to resolve the issues mentioned in the issues faced section (Main Menu & Pre-Legal Seniors classified as non-senior legal issues, questionable outcome results of OtherLegalMenu, and differentiating between Legal Menu 1 and 2). It could also be helpful to try and eliminate the 'Unknown' subtypes as this was the default option so finding the source of these could help reveal where there are gaps or inconsistencies in the tagging.

- References
    - Utilized the call_type_tag code to add the high-level groupings to the dataset https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/caller_type_tag_1Dec_Krish.ipynb

Presentation 4

- Links to presentation(s) and code(s) on GitHub

Code:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/CallOutcomesClassification_19Nov_Vivienne.ipynb

Presentation:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/OutcomeTagging_Nov18_EDASimple2.pdf

Excel:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Clarifications/CAR_data_Activity_name_tagging_10Nov.xlsx

Video:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/VideoLink_Nov18_EDASimple2.txt

- What did you do?

For this presentation, our group's objective was to classify each activity, present in the dataset as a call journey's final activity, as positive, negative, or neutral and create a rule for defining these in an Excel sheet before implementing it in code to find the overall distribution of outcomes in the dataset as well as provide the distribution of outcomes for each menu. We opted to divide the classifying and verifying, and actual tagging of calls into two distinct tasks since our group is split in terms of preference of langage between R and Python, and we felt it would be simpler to aggregate all the outcomes, particularly as many activities are found across several menus if all the tagging was done in one language. As such, I focused on creating the initial tag for activities, Sabrina focused on verifying the classifications, and then Caroline, Hailee, and Nikole worked on tagging all of the outcomes in R and creating visualizations.

In classifying the outcomes, I first checked the flow charts to see if the activity was present in any menus and looked to see if the flow of whichever menus it was present in suggested that it should be the final step of a call (i.e. terminal node). If it was a terminal node, I then looked more closely at the verbiage and the purpose of the activity to decide if there needed to be an additional rule to determine if it was positive, negative, or neutral (e.g. there needs to be an agent name for a call to be considered positive). If it did not initially appear to be a terminal node, I looked at the verbiage to see if it provided an explanation as to why a call may end at that point (e.g. a menu telling the caller that LAC can only serve issues within Cook County so they do not make a selection and progress in their call), then I used that information (or lack thereof) to create a rule for classifying the activity.

For activity names that did not correspond to a step in the flow chart, they seemed to mainly be internal steps of the phone system so I used a combination of inspection of examples within the dataset and generative AI to help interpret the call journeys and their outcomes. In looking at the dataset, I both filtered for contact session IDs that contained the activity name at any point in their journey and I looked specifically for cases where the activity was the call's final activity. I generally used those two approaches to get a group of contact session IDs and then looked at the full call journey for a sampling of each kind of journey. I also asked generative AI to help explain call journeys and outcomes where there was ambiguity or where it was difficult to interpret the activity's meaning based on its name alone. Through these approaches, I was able to assign classifications to a majority of the activities, and I feel there are no recommendations based on this work since it was mainly internal but there is still the need for some additional clarity to refine the classifications.

- How does it help the project?

This helps the project by defining rules for classifying outcomes that are not based on termination reason so that the classifications can be extended to all available data to provide better insights about call outcomes overall and within specific menus.

- Issues faced (if any)

An issue faced was that there were certain steps that were unclear in how they related to the overall state of the call and there did not seem to be any additional information to help clarify. In particular, 'LegalServerScreenPop' and 'ScreenPopProcessComplete' were difficult to interpret because in my understanding, they are internal steps while an agent is being attached to a call, but this happens prior to the customer being connected (i.e. before it dials or as it is ringing) so there will be an agent name associated with the call after this point but there is not a clear indicator of once the customer has been reached or joined the call (agent could wait for call as it rings). As an initial rule, I considered these to be positive if there was an agent name and a duration after either activity but I was unsure of what lower limit should be placed on call length to be sure the customer was actually reached and I felt that there should be a more rigorous rule in place. Additionally, for rows that do have termination reasons, it was also seen that sometimes there are rows of activity even after the termination reason, and these were often several minutes after (i.e. not separate rows reflecting two sides of simulataneous action) so this also raised doubts about the rule's effectiveness. I was more generally confused about the relationship between the two activity reasons as it seems that 'LegalServerScreenPop' should have to be followed by 'ScreenPopProcessComplete' in order to be successful, but it also seemed unlikely that all calls that ended with 'LegalServerScreenPop' were unsuccessful and in the explanations of call journeys from the examples I provided to generative AI, it seemed to suggest there were cases where a successfully connected call could have 'LegalServerScreenPop.' Still, I was wary of the accuracy of these claims but I struggled to find information from the phone system itself about the meaning of these activity names so I was unable to verify the explanations. I also chose neutral as the other outcome for 'ScreenPopProcessComplete' and 'LegalServerScreenPop' because I felt that if the call was not connected, that was due to the customer being unreachable. I was unsure if there was a scenario where these would actually be considered unsuccessful courtesy callbacks, but I did not know how to distinguish between these scenarios.

Similarly for 'GetLoggedIn<*>Agents,' it was difficult to rely on the call journeys themselves for insight but I was also hesitant to just accept the interpretation from generative AI as it appeared to be an activity name that was more tailored to Legal Aid Chicago than a standard system term. Based on the explanation, it seemed to be a system issue where the phoen system should check if there are agents logged into a specific queue and then the system decides whether to connect a caller or place them in a queue, but instead, it had just stalled at that step for an unknown reason. There seemed to be two cases where the step and the end of the call occurred simultaneously but sometimes the order was flipped (see examples in code) where either the activity name or a NaN would be last and it seemed dependent on whether there was a termination reason for that call. From this, it seems difficult to distinguish if the end of the call was initiated by the customer without using termination reason, especially if there is a difference based on the order of the NaN activity and the 'GetLoggedIn<*>Agents' (i.e.

one kind is the system and one is the cusomer) so I feel this needs further clarification of the activity's meaning and of whether these cases are just a quirk that developed as the system got updated to include termination reasons.

For the 'No activity' term, I was unsure of where this activity name had come from as I did not see it as a value anywhere in the dataset but I was then unsure of how it had been measured it was just reflecting that there was no activity because it seemed that many other steps related to a lack of activity but were still considered with their previous activity name.

For 'TransferToSafeHaven,' from other team's work and discussion, it was understood to be the trial transfer to the chatbot, but there was not enough information about how callers were told to select that option or how the call's were actually handled to consider an outcome.

- Attempts to resolve issues (if any)

A minor issue was the lack of verbiage for certain prompts, in particular the voicemail transfers, as it was not initially possible to find the outcomes based on our existing materials, but these were easily resolved by calling Legal Aid. However, not all were as straightforward as there are activity names that seem to correspond places where a caller would hear a message but it is not as clear what journey gets there, for example, 'SubSeniorPreQueueMessage1_1' which I assumed to fill the same position as 'PreQueueMessage1' only with a distinction that it should play for the senior queues but I had difficulty navigating when calling legal aid as I tried to use the flowcharts as a reference but they no longer matched up with the call flow, and for certain steps there is no specific indication of what it should correspond to.

- Issues resolved (if any)

I realized during my classifying that I had been inconsistent with my identification of the 'Last Menu' as I had for calls that ended with an activity that was related to a queue or courtesy callback that I had been identifying them with their flow chart (i.e. 'Queues' or 'Courtesy Callbacks') instead of indicating all the menus that could possibly have calls enter a queue or receive a courtesy callback. I updated the Excel sheet so that these activity names now have all possible menus that could lead to them as their last menu options.

- Next Steps

  Create and refine the rules for activity names where there was ambiguity and then help verify how they were implemented in the code.

- References (mention if you built up on someone else's work)
  - Import file for CAR dataset https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/CAR_data_import_code.ipynb

Presentation 3

- Links to presentation(s) and code(s) on GitHub

Code:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/LegalMenuOutcomes_5Nov_Vivienne.ipynb

Presentation:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/CallMenuOutcomes_4Nov_EDATeam2.pdf

- Presentation does not use waterfall transitions for slides with diagrams to simplify presentation

Presentation Video:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/VideoLink_Nov5_EDASimple2.txt

Document for Outcomes & Rules

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/OutcomesRulesDoc_Nov5_EDASimple2.pdf

- What did you do?

For this presentation, my objective was to quantify the number of calls that had been identified as having positive, negative, or neutral outcomes, as well as document the rules and outcomes that were being considered to define these categories. Our team decided to focus on the five most popular menus with each group member being responsible for identifying the outcomes for a menu. I worked on the Legal Menu, mostly focusing specifically on LegalMenu2 and I looked only at pathways that had their terminal node within LegalMenu2 (as many of the menu options bring the user to additional menus so their outcomes will be identified by the menu of their terminal node). I began by filtering the overall CAR dataset to only include contact session IDs that had 'LegalMenu1' as an activity name in one of their rows, then I utilized the same data cleaning methods as for presentation 2 (adding a column for duration, time difference between steps of the call journey, and a call ID, again utilizing Meera's data cleaning work). The pathways that could enter their terminal node directly from LegalMenu2 were ADAPT, Consumer, Criminal Records, and from LegalMenu1, the only other pathway than LegalMenu2 is to a Front Desk Transfer (to get an interpreter for a non-English or non-Spanish speaking caller). For the ADAPT and Consumer options, selecting either option enters the caller into the respective queues and for Criminal Records, selecting the option leads the caller to listen to a recorded message indicating further action, but does not allow the caller to speak to an agent. For the Front Desk Transfer, after checking if the call is during open hours, the caller is connected to the front desk or the queue if there is no one available.

Based on this, I considered an approach that could be applied to ADAPT, Consumer, and Front Desk Transfer as they all dealt with scenarios where the caller would be entered into a queue. Once the caller is in a queue, a positive outcome would mean that they are able to speak to an agent about their issue, either directly from their original call or through a courtesy callback. I did not make a distinction between these two scenarios at this point as the call outcome is the same regardless and the same contact session ID would be used to identify the call at either stage. To identify the calls that entered into each queue, I filtered for contact session IDs where there was an activity name that reflected the caller selecting the queue ('ADAPTQueue,' 'ConsumerQueue,' 'FrontDeskTransfer'). As there seems to be no way to go back to an earlier step in the journey once a caller has entered a queue, I thought this would be sufficient to ensure that it was the final step in a call, but it may require further exploration to confirm. The two criteria used to check if the call had successfully reached an agent was that the termination reason for the call was agent left, or the termination was customer left and that row also had an agent name. I used a mask to turn this into a function that returned an array of the unique contact session IDs in the dataframe that met this criteria, and used the length of this array to identify the number of calls with a positive outcome. Then, I filtered the dataframe to exclude the contact session IDs of the calls with positive outcomes.

In some cases, the final row has an agent name but no termination reason and the second to last row has termination reason, ‘Agent Left’ but no agent name (see example below). I considered that these to still be successful calls since this seemed to just be an odd representation of the agent ending a call. These calls should be identified as successful, however, I wanted to protect against cases where there is a similar anomaly of there being no termination reason in the final row, but the call outcome should be neutral or negative. For this, I took the unsuccessful call dataframe and then grouped the calls by call ID and assigned a count to each row in descending order(so the final row for the call would be considered 0), and checked for termination reasons that were NaNs. Then, I filtered the unsuccessful calls dataframe to exclude rows that were the final row for their call ID and did not have a termination reason. To find the negative outcomes, I grouped this filtered dataframe by call ID, looking at each call’s last termination reason, and then checked if the termination was in a list of negative termination reasons. For calls that entered a queue, I considered that there was not a neutral outcome because if they reached this point, LAC must be open and there must be a queue for their issue based on their selection. For the remaining calls, I considered that these must be outcomes that could not be captured by the scenarios I had included for positive or negative, and I found these by comparing the length of the non-positive outcome dataframe to the negative outcome dataframe length.

528371	26660	90e15653-88a6-451c-8bdf-f1add5ac3043	All LAC Queues Telephony EP	NaN	ADAPTQueue	NaN	NaN	NaN	2025-08-27 15:07:57	57.0	0 days 00:32:09
528372	26660	90e15653-88a6-451c-8bdf-f1add5ac3043	NaN	NaN	NaN	ADAPT	NaN	NaN	2025-08-27 15:07:57	0.0	0 days 00:32:09
528373	26660	90e15653-88a6-451c-8bdf-f1add5ac3043	All LAC Queues Telephony EP	NaN	NaN	NaN	NaN	NaN	2025-08-27 15:07:57	0.0	0 days 00:32:09
528374	26660	90e15653-88a6-451c-8bdf-f1add5ac3043	All LAC Queues Telephony EP	NaN	NaN	ADAPT	NaN	NaN	2025-08-27 15:07:57	0.0	0 days 00:32:09
528375	26660	90e15653-88a6-451c-8bdf-f1add5ac3043	All LAC Queues Telephony EP	NaN	PreQueueMessage2	NaN	NaN	NaN	2025-08-27 15:07:57	0.0	0 days 00:32:09
528376	26660	90e15653-88a6-451c-8bdf-f1add5ac3043	All LAC Queues Telephony EP	NaN	PlayMOH300s	NaN	NaN	NaN	2025-08-27 15:08:09	12.0	0 days 00:32:09
528400	26660	90e15653-88a6-451c-8bdf-f1add5ac3043	All LAC Queues Telephony EP	NaN	QueueMenu1	NaN	NaN	NaN	2025-08-27 15:13:10	301.0	0 days 00:32:09
528401	26660	90e15653-88a6-451c-8bdf-f1add5ac3043	All LAC Queues Telephony EP	NaN	PlayMOH300s	NaN	NaN	NaN	2025-08-27 15:13:31	21.0	0 days 00:32:09
528457	26660	90e15653-88a6-451c-8bdf-f1add5ac3043	All LAC Queues Telephony EP	NaN	QueueMenu1	NaN	NaN	NaN	2025-08-27 15:18:32	301.0	0 days 00:32:09
528459	26660	90e15653-88a6-451c-8bdf-f1add5ac3043	All LAC Queues Telephony EP	NaN	PlayMOH300s	NaN	NaN	NaN	2025-08-27 15:18:54	22.0	0 days 00:32:09
528477	26660	90e15653-88a6-451c-8bdf-f1add5ac3043	All LAC Queues Telephony EP	NaN	QueueMenu1	NaN	NaN	NaN	2025-08-27 15:23:54	300.0	0 days 00:32:09
528478	26660	90e15653-88a6-451c-8bdf-f1add5ac3043	NaN	CourtesyCallback	NaN	NaN	NaN	NaN	2025-08-27 15:24:13	19.0	0 days 00:32:09
528479	26660	90e15653-88a6-451c-8bdf-f1add5ac3043	Courtesy Callback Telephony EP	NaN	ReadANI	NaN	NaN	NaN	2025-08-27 15:24:13	0.0	0 days 00:32:09
528480	26660	90e15653-88a6-451c-8bdf-f1add5ac3043	Courtesy Callback Telephony EP	NaN	NaN	ADAPT	Amanda Palacios	NaN	2025-08-27 15:24:17	4.0	0 days 00:32:09
528481	26660	90e15653-88a6-451c-8bdf-f1add5ac3043	Courtesy Callback Telephony EP	NaN	LegalServerScreenPop	NaN	NaN	NaN	2025-08-27 15:24:17	0.0	0 days 00:32:09
528482	26660	90e15653-88a6-451c-8bdf-f1add5ac3043	Courtesy Callback Telephony EP	NaN	NaN	ADAPT	Amanda Palacios	NaN	2025-08-27 15:24:18	1.0	0 days 00:32:09
528539	26660	90e15653-88a6-451c-8bdf-f1add5ac3043	NaN	NaN	NaN	ADAPT	Amanda Palacios	NaN	2025-08-27 15:33:21	543.0	0 days 00:32:09
528540	26660	90e15653-88a6-451c-8bdf-f1add5ac3043	Courtesy Callback Telephony EP	NaN	NaN	ADAPT	NaN	Agent Left	2025-08-27 15:33:21	0.0	0 days 00:32:09
528555	26660	90e15653-88a6-451c-8bdf-f1add5ac3043	NaN	NaN	NaN	ADAPT	Amanda Palacios	NaN	2025-08-27 15:35:21	120.0	0 days 00:32:09

For the Criminal Records, the call journey once the option has been selected from LegalMenu2 involves being entered into a queue then immediately transferred to listen to a message about next steps for callers looking to have criminal records sealed or expungement. In this case, I considered ‘Agent Left’ to be a positive termination reason because the call automatically hangs up once the full message has been played so this indicates that the caller listened to the entirety of the message. I considered ‘Customer Left’ to be a neutral termination in this case because the caller was able to reach the correct endpoint for their journey, but they may miss out on information necessary to get help from LAC or they could recognize that LAC’s services do not actually fit their needs. In either case, these reasons for the call termination are unrelated to the call journey and are out of LAC’s control, but still represent callers who may not receive all necessary information.

Finally, I included the value counts for the termination reasons for the negative outcomes for each menu in case this would be helpful for future work identifying the drivers behind negative outcomes for each menu. Then I aggregated the number and proportion of positive, negative, neutral, or unexplained for the overall legal menu, and then focused specifically on those that terminate within the Legal Menu.

- How does it help the project?

This helps the project by beginning to identify outcomes by menu to help Legal Aid Chicago identify which menus and call paths ways are having success or trouble in addressing customer legal issues. This work will hopefully allow for larger trends to be identified in what factors contribute to successful and unsuccessful call outcomes so changes can be made to help improve the efficacy of the different menus.

- Issues faced (if any)

An issue faced in this analysis was the inability to categorize the outcomes for FrontDeskTransfer properly in the context of this prompt's actual purpose. Based on the verbiage, there is an option in LegalMenu1 for the caller to press '*' to request an interpreter, and this is the only selection that a caller can make at this point (if the caller makes no selection, they proceed to LegalMenu2). I'm concerned that there would not be a clear way to differentiate between callers who select this because they need an interpreter versus callers who select it because it is the only select option available at that point or because they are trying to get a quicker path to an agent. I feel that callers just trying to reach an agent may be deterred by the fact that the verbiage does not refer directly to being transferred to the front desk or an agent, but I am also unsure if there are callers who may call multiple times as they try to find a quicker pathway. This could potentially be a question to ask Cynthia as she would likely have better information based on the agents' experience of the types of calls that reach the front desk through this pathway.

I was also unsure if 'Customer Left' should be considered a neutral termination reason for Criminal Records because I believe the only possible termination reasons for that path were 'Agent Left,' 'Customer Left,' or a reason that references a system error so I was unsure how informative the analysis would be if it considered an outcome besides a system error positive. However, I was unsure if considering 'Customer Left' neutral when the phone menu had in a large part worked as intended should likely be considered positive.

Additionally, I ran into some issues with certain calls that had inconsistencies in termination reasons that I was unsure if they could potentially cause issues in identifying outcomes, either due to double counting or misrepresenting outcomes.

601811	30451	8606a56f-b44d-4867-800a-363834552ce6	NaN	CourtesyCallback	NaN	NaN	NaN	NaN	2025-09-12 08:06:46	3.0	0 days 21:01:35
601812	30451	8606a56f-b44d-4867-800a-363834552ce6	Courtesy Callback Telephony EP	NaN	ReadANI	NaN	NaN	NaN	2025-09-12 08:06:46	0.0	0 days 21:01:35
601813	30451	8606a56f-b44d-4867-800a-363834552ce6	Courtesy Callback Telephony EP	NaN	CollectCallbackNumber	NaN	NaN	NaN	2025-09-12 08:06:49	3.0	0 days 21:01:35
601818	30451	8606a56f-b44d-4867-800a-363834552ce6	Courtesy Callback Telephony EP	NaN	CollectCallbackNumber	NaN	NaN	NaN	2025-09-12 08:06:54	5.0	0 days 21:01:35
601824	30451	8606a56f-b44d-4867-800a-363834552ce6	Courtesy Callback Telephony EP	NaN	CollectCallbackNumber	NaN	NaN	NaN	2025-09-12 08:07:01	7.0	0 days 21:01:35
601840	30451	8606a56f-b44d-4867-800a-363834552ce6	Courtesy Callback Telephony EP	NaN	ConfirmCallbackNumber	NaN	NaN	NaN	2025-09-12 08:07:12	11.0	0 days 21:01:35
601854	30451	8606a56f-b44d-4867-800a-363834552ce6	Courtesy Callback Telephony EP	NaN	CCB	NaN	NaN	NaN	2025-09-12 08:07:23	11.0	0 days 21:01:35
601855	30451	8606a56f-b44d-4867-800a-363834552ce6	NaN	NaN	NaN	ADAPT	NaN	NaN	2025-09-12 08:07:23	0.0	0 days 21:01:35
601856	30451	8606a56f-b44d-4867-800a-363834552ce6	Courtesy Callback Telephony EP	NaN	PlayCCBConfirmation	NaN	NaN	NaN	2025-09-12 08:07:23	0.0	0 days 21:01:35
606191	30451	8606a56f-b44d-4867-800a-363834552ce6	NaN	NaN	NaN	ADAPT	NaN	Queue Timeout	2025-09-13 05:06:43	75560.0	0 days 21:01:35
606192	30451	8606a56f-b44d-4867-800a-363834552ce6	NaN	NaN	NaN	ADAPT	NaN	Customer Left	2025-09-13 05:06:43	0.0	0 days 21:01:35
606193	30451	8606a56f-b44d-4867-800a-363834552ce6	Courtesy Callback Telephony EP	NaN	CallbackRetry	NaN	NaN	NaN	2025-09-13 05:06:53	10.0	0 days 21:01:35
606194	30451	8606a56f-b44d-4867-800a-363834552ce6	NaN	NaN	NaN	ADAPT	NaN	NaN	2025-09-13 05:06:53	0.0	0 days 21:01:35
606195	30451	8606a56f-b44d-4867-800a-363834552ce6	NaN	NaN	NaN	ADAPT	NaN	Customer Left	2025-09-13 05:06:53	0.0	0 days 21:01:35
606196	30451	8606a56f-b44d-4867-800a-363834552ce6	Courtesy Callback Telephony EP	NaN	NaN	ADAPT	NaN	Customer Left	2025-09-13 05:06:54	1.0	0 days 21:01:35

In this example, it appears that the caller booked a courtesy callback but the call is not shown to have a termination reason, and the following row is roughly 20 hours later with the termination reason 'Queue Timeout' so I'm unsure if this is suggesting that the caller somehow remained in the queue the whole time, despite booking a callback. The row following the 'Queue Timeout' also has the same timestamp and also has the termination reason 'Customer Left. Then it appears that there are two callback retries that are both terminated by the customer within a second of each other. I believe that this example should not cause any issue because all of the termination reasons are negative and looking at distinct call IDs should hopefully avoid double counting, but it still raises issues of how to ensure that the identification of outcomes is effective even for irregular cases.

As my analysis for this presentation focused on call outcomes, I mainly targeted my work towards calls that reached a terminal node so I believe that I will need to do further work to incorporate checks that can account for call outcome for calls that terminate without entering a queue or advancing in the menu (e.g. in this case, there was 392 calls that entered Legal Menu 1, but never made it into Legal Menu 2).

Finally, it seemed that the proportion of positive outcomes was fairly high and I was unsure if this was indicative of my criteria for identifying positive outcomes not being rigous enough. Still, though there is a significant number of calls that pass through the Legal Menu, the menus that terminate within it have a relatively small number of calls so it is possible that they have better outcomes due to the smaller volume of callers attempting to access those queues. I feel that this may need further verification, possibly through being checked by another group member.

- Attempts to resolve issues (if any)

  Addressed in issues resolved section

- Issues resolved (if any)

  As mentioned in the first section, I believe that removing any final row for a call (looking at negative or neutral outcopmes) that lacks a termination reason will help ensure that it can effectively identify the correct termination reason and call outcome for calls based on the cases I was able to identify in my analysis. I also recognize though that there are a variety of representations of actions in the data so this may require further refining

- Next steps

  For next steps, I would like to check that all calls identified as moving through Legal Menu to other menus actually continue on their journey, and to develop a criteria to formally assign an outcome for calls that were identified as not progressing to another menu in this analysis. I also think it would be worthwhile to have our group work to audit each of our overall logic and specifics of code in classifying calls up until this point.

- References (mention if you built up on someone else's work)
  - Import file for CAR dataset

    https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/CAR_data_import_code.ipynb

    - Meera’s data cleaning
      https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Verbiage%20analysis/Meera%20Trivedi/Customer%20Abandonment_20Oct_Meera.ipynb

Presentation 2

- Links to presentation(s) and code(s) on GitHub

Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/CallCategorization_Oct23_Group2.pdf
Code:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/IdentifyingAbandonedCalls_23Oct_Vivienne.ipynb

- What did you do?

For this presentation, my objective was to identify scenarios where call outcomes can be categorized as positive, negative, or neutral so that eventually the activity as the customer leaves can be visualized. To begin this, I looked at possible values for termination reasons included in the dataset and tried to identify which could be easily mapped to the three outcome categories. Then, I tried to identify call outcomes that either would not be reflected in the termination reason (e.g. for a courtesy callback, it is likely that the customer left their original call after selecting courtesy callback, but there then could be several attempts to reach them before they successfully speak to an agent so there would be several termination reasons for one ‘Contact Session ID’ and categorizing the call outcome would need to incorporate additional factors). Based on these scenarios, I then began to try and identify what additional factors would be needed to categorize the outcomes that did not match exactly with a termination reason. In particular, ‘Customer Left’ appeared to be the termination reason that had the most scenarios that needed further exploration. To begin this, I chose to focus on abandoned calls which I considered to be negative outcomes where the termination reason was ‘Customer Left,’ and I developed additional scenarios specifically for abandoned calls, building on the cases that had been identified in our mentor meeting and the examples proposed by Meera and Sana (Simple EDA Team 1). After identifying the different cases, I began trying to find a method identifying these scenarios in the dataset that can hopefully later be applied to visualizing the different outcomes. Since it appeared that the duration of each step in the journey and the overall call would be helpful in identifying if the call was abandoned, using the CAR dataset filtered to only include ‘Contact Session IDs’ where the termination reason was ‘Customer Left,’ I added a duration column and also incorporated Meera’s data processing that added a ‘Time Difference’ column for each step of a caller’s journey and the ‘Call ID’ column.

The first case I considered was the call being abandoned while the hold music was playing. I began by filtering my dataset to only include ‘Contact Session IDs’ that had ‘PlayMOH300s’ as the ‘Activity Name’ in at least one of its rows. I then sorted the dataframe by ‘Call ID’ and ‘Activity Start Timestamp’ so all rows for each call would be successive in the dataframe according to the time of each step. An example call from this dataframe is included in my code for this presentation. To find cases where the caller left during the hold music, my methodology was to identify cases where listening to hold music was in the row immediate prior to the call termination (i.e. it would have to be the caller’s last activity so their issue could not have been resolved) and the ‘Time Difference’ between the last two steps was less than 300 seconds so they did not listen to the music in its entirety. I first wrote an ‘if’ statement based on these two conditions and utilized the index of an example I had identified from the dataframe to test my logic, and then I extended this to a function that could be applied to the dataframe as a whole with assistance from AI. From this, I saw that roughly 10% of calls were abandoned during the hold music out of the total abandoned calls that listened to hold music. Though not a focus of this analysis, I felt it was a chance to check that the values were reasonable and help understand if the scenarios I had identified were making up the majority of abandoned calls. Based on the suggestion in our mentor meeting, I checked if there were any calls that had an activity in their final row by filtering to only include the final row of the calls in the broader dataframe and then filtered the resulting dataframe for any values where there was an activity name. There turned out to be no rows which fit this criteria so this may help eliminate one possible abandonment scenario. From this, the next scenario I tried to identify was abandoned calls that never enter a queue, though at this point, I was not yet able to make the distinction between very brief calls that never enter a queue (e.g. LAC is closed so they should not be able to get in a queue; caller immediately hangs up) which appears to be neutral versus longer calls where the caller takes many steps but cannot find the proper menu and queue. To begin this, I found an example by finding the ‘Call ID’ value with the maximum number of instances as I thought this would reflect the call with the most steps in its journey, and I thought that many steps would indicate the user was unable to find the proper menu. Upon inspection, this example was an abandoned call where the caller seemed to rotate between Housing, Tenant, and PreTenant menus for roughly two hours. In terms of identifying abandoned calls that did not enter queues, I grouped the dataframe by ‘Call ID’ and applied a lambda function to check if the ‘Queue Name’ for all rows was NaN, then since I wanted to preserve the additional information in the dataframe, I took the ‘Call IDs’ from the grouped dataframe and used them to filter the original dataframe and found the number of calls by counting the unique Call IDs. There were 45,396 abandoned calls that never entered a queue which was significant compared to the 69,154 with the termination reason ‘Customer Left,’ but without further exploration, it cannot be said that these are all negative outcomes.

- How does it help the project?

This helps the project by developing a method to identify different categories of outcomes from the data, and works towards the goal of visualizing the different call outcomes by activity type so hopefully Legal Aid Chicago can better understand the factors that contribute to successful and unsuccessful call outcomes.

- Issues faced (if any)

There were not any major issues in this analysis, but I think as I continue working on this task, there will be the possibility for issues to arise if my set of scenarios for outcomes does not cover all possibilities, as well as the difficulty in possibly identifying where these gaps are within the data. Additionally, I began to run into the issue with identifying abandoned calls that never enter a queue because I was unsure of how to construct a criteria that could cover all possibilities as well as how granular my distinctions should get with regard to the different scenarios. For example, I did not see a clear metric for a differentiating between a short (neutral) call that does not enter a queue versus a long (negative) call that does not enter a queue, and I was not sure if a certain number of steps or duration of call would be a strong solution to this issue. I was also unsure about how detailed to go in identifying different scenarios as I know the goal is the three broad categorizations, but I wasn't sure if this should be balanced against including more detail about the activity for the visualization. Smaller issues included being unable to assign certain termination reasons to positive, negative, or neutral because I was unsure of their meaning ('RONA_TIMER_EXPIRED,' 'RONA Timer Expired,' 'CONTACT_CALLBACK_IN_PROGRESS'). However, they all seemed to have relatively few instances in the data so I felt okay in that they would not massively affect my analysis at this time (exact values are included at the very end of my code).

- Attempts to resolve issues (if any)

To resolve the difference between neutral and negative calls abandoned without entering a queue, I had initially tried to limit the duration of a call to a minute, assuming that I would adjust it depending on the trends I saw in the initial results. However, this had revealed that there were calls that were handled effectively (e.g. LAC closed and the user listens to the message then leaves) so I was unsure if this could miss out on insights based on the difference between the situations. For example, if a caller is so overwhelmed by hearing the initial message or attempting to begin navigating through the menus that they would rather hang up, I feel that is a negative outcome but I am unsure of if this can effectively be identified from the dataset without focusing too closely on any one example.

- Issues resolved (if any)

By adding Call ID and then sorting by it and Activity Start Timestamp, it resolved the issue of not being able to utilize consecutive dataframe rows to track the progression of a call's steps as previously successive steps of a call could be separated by any number of rows depending on the other calls occurring concurrently and their activity. This was useful in the development of a method to find calls that were abandoned while listening to the hold music. This further helped address issues of wanting to be able to group all entries of a call together without losing any information by using .groupby(). Additionally, I originally had issues with my function to find calls abandoned during the hold music as the time difference for the first row of each call was NaN, but this was easily resolved by filling these values with zero.

- Next steps

For next steps, I will refine the criteria used to identify the different categories of calls abandoned without entering a queue, and continue to develop methods for identifying the remaining abandonment scenarios from the data. I would also like to try and incorporate call outcome categorization or reason into the dataframe as a column as I feel it could help support visualization.

- References (mention if you built up on someone else's work)
    - Import file for CAR dataset
      https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/CAR_data_import_code.ipynb

    - Meera's data cleaning
      https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Verbiage%20analysis/Meera%20Trivedi/Customer%20Abandonment_20Oct_Meera.ipynb

Presentation 1

- Links to presentation(s) and code(s) on GitHub

Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Data%20integrity%20checks/CARAllCallsIntegrity_7Oct_EDATeam2.pdf

Code:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/CallOutcomeAggregation_7Oct_Vivienne.ipynb

- What did you do?

My focus for this presentation was beginning to see if the CAR dataset could be utilized to replicate the Legal Menu Summary's information about call outcomes so that future analysis could rely only on the CAR dataset. I began by using the 'CAR_import_data' file to read in the data, and I also specifically created a folder that only contained the months where the CAR dataset included 'Termination Reason' as a column. I began by looking at the unique values for 'EP Name,' 'Queue Name' and 'Flow Name' to try to gauge what each step represented. My initial approach to tracking the call journey (in order to approximate which menu they had selected and which queue they entered; shown in Legal Summary Menu through the 'Legal Menu Option,' 'Menu Selection,' 'Queue Selection,' and 'Group suboption' columns) was to use the combinations of 'EP Name' and 'Queue Name' but I recognize this as a rough method until I can develop a more nuanced understanding of how the CAR data represents the call pathways.

My first step was to create a dataframe with the total calls for each menu and queue combination by grouping by 'EP Name' and 'Queue Name' and then aggregating by unique 'Contact Session ID' values. I also filled the NaNs in the 'EP Name' and 'Queue Name' columns with 'Unknown' so that if there was a significant number of calls missing one of these columns, it could hopefully be used to help identify in which step or queue was causing issues. This was intended to begin recreating the 'Total Calls' column in Legal Menu Summary and also provide a baseline for comparison as I aggregated other call outcomes. For abandoned calls, I began by creating a subgrouping of the CAR dataset that only includes the periods where 'Termination Reason' was included as a column in the data as this creates the basis for determining the call outcome. I filtered the dataframe to only include the last instance of each 'Contact Session ID' as it seemed that this would be the best way to identify unique calls and the final step of the caller's journey. However, I do recognize that there are potentially issues with this method depending on how call outcomes are being defined but I will discuss this further in the 'Issues faced' section. For open queue calls, I filtered the dataset for rows that had a 'Queue Name' value and did not have a 'Termination Reason' as I reasoned this would reflect ongoing calls where the caller was in a queue. The same method for aggregating total calls was applied to abandoned and open queue calls.

- How does it help the project?

This work helps the project because it helps further the goal of being able to conduct analysis on call outcomes while working to streamline the process by only utilizing one dataset. Additionally, it seems that the aggregation methods behind the Legal Menu Summary dataset are unclear so having an aggregation where the methods are known could help facilitate further analysis about trends in call outcomes.

- Issues faced (if any)

The main issue faced at this stage in analysis is understanding how these journeys through the different menus and queues are reflected in each dataset, and then how to effectively use that to replicate the summarized information from the Legal Menu Summary dataset with the CAR dataset. This also raised issues of how to define different outcomes based on the values available in the dataset, for example, ‘Customer Left’ could be taken as the customer abandoning their call, but it could also represent that they spoke to an agent and were successful with getting help with their legal issue. At this time, I felt I did not have enough familiarity with the dataset and the call flow to effectively define these outcomes through a combination of values. Additionally, there seems to be certain issues with the data itself and how it is being compiled or how the phone system itself is interpreting different actions. In the image below, it can be seen that the customer navigates through the menu and eventually selects receiving a courtesy callback about their issue. After several attempts to reach the customer, it seems that the agent and customer are both able to get on the call. However, after the customer hangs up, there are two more entries for the call (one at the same time as the customer is shown leaving, and the other two minutes later) but neither has an ‘EP Name’ or a ‘Termination Reason’ so if these were looked at as the last step in the call journey, all information about the call back and why the call ended would be lost. I recognize this issue is partially as result of my approach of only looking at the last instance of each ‘Contact Session ID,’ however, I’m unsure of how to systematically address this issue if it is a larger issue with nonconsecutive entries for a single call or if I just need a better understanding of the procedure used by the phone system to track and compile the steps in the journey.

492	3e0c8f9b-f617-4444-8b74-63d829f18c98	NaN	NaN	NaN	2025-03-17 08:03:32	Family	NaN	NaN	8
493	3e0c8f9b-f617-4444-8b74-63d829f18c98	All LAC Queues Telephony EP	NaN	NaN	2025-03-17 08:03:32	NaN	NaN	NaN	8
494	3e0c8f9b-f617-4444-8b74-63d829f18c98	All LAC Queues Telephony EP	NaN	NaN	2025-03-17 08:03:32	Family	NaN	NaN	8
495	3e0c8f9b-f617-4444-8b74-63d829f18c98	All LAC Queues Telephony EP	NaN	PreQueueMessage2	2025-03-17 08:03:32	NaN	NaN	NaN	8
496	3e0c8f9b-f617-4444-8b74-63d829f18c98	NaN	CourtesyCallback	NaN	2025-03-17 08:03:42	NaN	NaN	NaN	8
497	3e0c8f9b-f617-4444-8b74-63d829f18c98	Courtesy Callback Telephony EP	NaN	ReadANI	2025-03-17 08:03:42	NaN	NaN	NaN	8
506	3e0c8f9b-f617-4444-8b74-63d829f18c98	Courtesy Callback Telephony EP	NaN	CCB	2025-03-17 08:03:58	NaN	NaN	NaN	8
507	3e0c8f9b-f617-4444-8b74-63d829f18c98	NaN	NaN	NaN	2025-03-17 08:03:58	Family	NaN	NaN	8
508	3e0c8f9b-f617-4444-8b74-63d829f18c98	Courtesy Callback Telephony EP	NaN	PlayCCBConfirmation	2025-03-17 08:03:58	NaN	NaN	NaN	8
512	3e0c8f9b-f617-4444-8b74-63d829f18c98	Courtesy Callback Telephony EP	NaN	DisconnectContact1	2025-03-17 08:04:08	NaN	NaN	NaN	8
838	3e0c8f9b-f617-4444-8b74-63d829f18c98	Courtesy Callback Telephony EP	NaN	NaN	2025-03-17 08:14:20	Family	Marisol Guadarrama	Customer Left	8
840	3e0c8f9b-f617-4444-8b74-63d829f18c98	Courtesy Callback Telephony EP	NaN	LegalServerScreenPop	2025-03-17 08:14:21	NaN	NaN	NaN	8
849	3e0c8f9b-f617-4444-8b74-63d829f18c98	Courtesy Callback Telephony EP	NaN	NaN	2025-03-17 08:14:35	Family	Marisol Guadarrama	Customer Left	8
937	3e0c8f9b-f617-4444-8b74-63d829f18c98	Courtesy Callback Telephony EP	NaN	NaN	2025-03-17 08:21:06	Family	Marisol Guadarrama	Customer Left	8
938	3e0c8f9b-f617-4444-8b74-63d829f18c98	NaN	NaN	NaN	2025-03-17 08:21:06	Family	Marisol Guadarrama	NaN	8
970	3e0c8f9b-f617-4444-8b74-63d829f18c98	NaN	NaN	NaN	2025-03-17 08:23:06	Family	Marisol Guadarrama	NaN	8

I also apologize for modifying Objective 3 after our group's mentor meeting - it had not been my intention but as I was working, I just felt that it may be ineffective analysis as I was still unsure in my understanding of how to aggregate the CAR dataset so I felt it may be better approached in later work. I will be sure in the future to ensure that any potential changes to objectives are raised in the mentor meeting.

- Attempts to resolve issues (if any)

  At this point, I did not have any attempts to resolve the issues because I felt that my understanding of the dataset was not robust enough to effectively recognize and address the issues in the representation of the calls.

- Issues resolved (if any)

  I did not resolve any issues at this stage, but the confirmation that 'EP Name' refers to the entry point will help inform future work.

- Next steps

  For next steps, I would like to refine the approach used to define the menu-queue combinations to ensure that they are accurate to the pathways actually available to customers and the respective outcomes. I would also like to see if it is possible to create a systematic approach to identifying the last entry that is relevant to the outcome of interest, and then work to verify that the method is effective across all entries; this may involve further exploration of the data to identify if there are trends in how different types of entries appear or if it varies on a case-by-case basis.